Time series: Stochastic Processes

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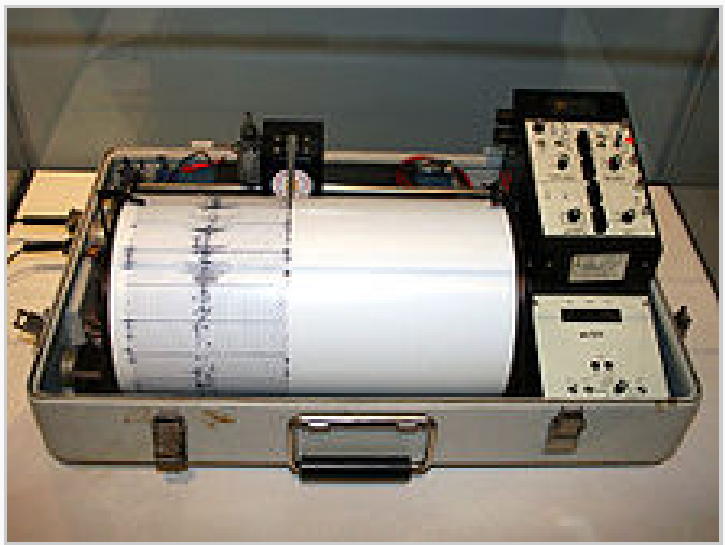
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Lecture notes 3

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Homework is given on the last slide

Features of time series, 1

A time series is said to be **continuous** when observations are made continuously in time.



A Kinemetrics seismograph, formerly used by the United States Department of the Interior.

A time series is said to be **discrete** when observations are taken only at specific times, usually equally spaced.

Features of time series, 2

In statistics, observations are usually **independent**,

that is, for a sample $(x_1, \ldots, x_N)$ we usually suppose that x_i and x_j are independent for all $i \neq j$.

In contrast, observations of a time series at different time points are typically **dependent**,

that is, for a time series $x_{t_1}, \ldots, x_{t_N}$ we typically have that x_{t_i} and x_{t_j} are not independent for all i, j.

Features of time series, 3

A time series is called **deterministic** if it can be predicted exactly.

A time series is called **stochastic** if the future values are only partly determined by past values, so that exact predictions are impossible.

Definition of stochastic process

A **stochastic process** is a family of time-indexed random variables

$$x_t(\omega),$$

where ω belongs to a sample space (Ω, P) and $t \in T$.

The set T is a discrete or continuous subset of $(-\infty, \infty)$.

For fixed t, $x_t(\omega)$ as a function of ω is a random variable.
For fixed ω, $x_t(\omega)$ as a function of t is called a **sample function** or **realization**.

A **time series** is a realization of a particular stochastic process.

Full description of a stochastic process

A stochastic process is fully defined by n-**dimensional cumulative distribution functions**

$$F_{t_1,\ldots,t_n}(u_1,\ldots,u_n) := P(\omega \ : \ x_{t_1}(\omega) \leq u_1,\ldots,x_{t_n}(\omega) \leq u_n)$$

for all $n \in \mathbb{N}$ and all $t_1,\ldots,t_n \in T$.

$F_{t_1,\ldots,t_n}(u_1,\ldots,u_n)$ is the cumulative distribution function of the vector $(x_{t_1},\ldots,x_{t_n})$.

For example, $F_{t_1,\ldots,t_n}(u_1,\ldots,u_n) = \prod_{i=1}^{n} \Phi_{t_i}(u_i)$, where $\Phi_{t_i}(u_i)$ is a standard c.d.f.

Stationary processes

A process is called **first-order stationary** if its one-dimensional distribution does not depend on a shift in time, i.e.

$$F_{t_1}(u_1) = F_{t_1+s}(u_1) \text{ for all } t_1, u_1, s.$$

A process is called **second-order stationary** if its two-dimensional distribution does not depend on a shift in time, i.e.

$$F_{t_1,t_2}(u_1, u_2) = F_{t_1+s,t_2+s}(u_1, u_2) \text{ for all } t_1, t_2, u_1, u_2, s.$$

A process is called n-**order stationary** if

$$F_{t_1,\ldots,t_n}(u_1, \ldots, u_n) = F_{t_1+s,\ldots,t_n+s}(u_1, \ldots, u_n)$$

for all $t_1, \ldots, t_n, u_1, \ldots, u_n, s$.

A process is called **strictly stationary** if it is n-order stationary for all $n \in \mathbb{N}$, $\mathbb{N} = \{1, 2, \ldots\}$.

Stationary processes, 2

If the n-dimensional c.d.f. $F_{t_1,\ldots,t_n}(u_1,\ldots,u_n)$ is sufficiently smooth, then there exists a n-**dimensional density** $p_{t_1,\ldots,t_n}(z_1,\ldots,z_n)$ such that

$$F_{t_1,\ldots,t_n}(u_1,\ldots,u_n) = \int_{-\infty}^{u_1} \cdots \int_{-\infty}^{u_n} p_{t_1,\ldots,t_n}(z_1,\ldots,z_n) dz_1 \ldots dz_n$$

and $p_{t_1,\ldots,t_n}(z_1,\ldots,z_n) \geq 0$.

Lemma 1. Higher-order stationarity of a stochastic process always implies lower-order stationarity.

Lemma 2. A process is n-order stationary if

$$p_{t_1,\ldots,t_n}(u_1,\ldots,u_n) = p_{t_1+s,\ldots,t_n+s}(u_1,\ldots,u_n)$$

for all $t_1,\ldots,t_n,u_1,\ldots,u_n,s$.

Characteristics of process

The **mean** function of a stochastic process $x_t = x_t(\omega)$ is

$$m(t) = \mathbf{E}x_t = \int_{-\infty}^{\infty} z p_t(z) dz,$$

$m(t)$ is a central (expected) value of the process at time t.

The **variance** function (if it exists) is

$$\sigma^2(t) = \mathrm{Var}(x_t) = \mathbf{E}(x_t - m(t))^2 = \int_{-\infty}^{\infty} (z - m(z))^2 p_t(z) dz,$$

$\sigma^2(t)$ is a measure of dispersion of x_t around $m(t)$.

Characteristics of process, 2

The **covariance** (autocovariance) function is

$$R(t,s) = \mathrm{Cov}(x_t, x_s) = \mathbf{E}[(x_t - m(t))(x_s - m(s))]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - m(z_1))(z_2 - m(z_2)) p_{t,s}(z_1, z_2) dz_1 dz_2$$

It is easy to see that $R(t,s) = \mathbf{E}[x_t x_s] - m(t)m(s)$.

The **correlation** (autocorrelation) function is

$$\rho(t,s) = \mathrm{Corr}(x_t, x_s) = \frac{R(t,s)}{\sqrt{\sigma^2(t)\sigma^2(s)}},$$

$\rho(t,s)$ is a measure of the linear dependence between x_t and x_s, $-1 \leq \rho(t,s) \leq 1$.

Stationarity in the wide sense

A stochastic process x_t satisfying $\mathbf{E}x_t^2 < \infty$ is called **(weakly) stationary in the wide sense** if the mean function $\mathbf{E}x_t$ and covariance $\mathrm{Cov}(x_t, x_{t+h})$ do not depend on t.

That is, $m(t) = \mathbf{E}x_t = m_0$ is a constant function and the covariance function $\mathrm{Cov}(x_t, x_s) = R(t,s) = R(t-s)$ depends only on the difference $t - s$.

Lemma 3. If x_t is a strictly stationary process and $\mathbf{E}x_t^2 < \infty$, show that x_t is stationary in the wide sense.

Lemma 4. If x_t is stat. in the wide sense, show that
a) $R(0) = \mathrm{Var}(x_t)$ for all t; b) $\rho(0) = 1$;
c) $|R(t)| \le R(0)$; d) $|\rho(t)| \le 1$;
e) $R(-t) = R(t)$ and $\rho(-t) = \rho(t)$.
Some proofs are given below.

Strict stat. and $\mathbf{E}x_t^2 < \infty$ implies wide-sense stat.

Since $F_t(z) = F_{t_0}(z)$ and therefore $p_t(z) = p_{t_0}(z)$ for all t, we have

$$m(t) = \mathbf{E}x_t = \int_{-\infty}^{\infty} z p_t(z) dz = \int_{-\infty}^{\infty} z p_{t_0}(z) dz = m(t_0).$$

Since $F_{t_1,t_2}(z_1, z_2) = F_{t_1+s,t_2+s}(z_1, z_2)$ and therefore $p_{t_1,t_2}(z_1, z_2) = p_{t_1+s,t_2+s}(z_1, z_2)$ for all t_1, t_2, s, we obtain

$$R(t_1, t_2) = \mathrm{Cov}(x_{t_1}, x_{t_2}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - m(z_1))(z_2 - m(z_2)) p_{t_1,t_2}(z_1, z_2) dz_1 dz$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - m(z_1))(z_2 - m(z_2)) p_{t_1+s,t_2+s}(z_1, z_2) dz_1 dz_2$$

$$\overset{[s=-t_2]}{=} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - m(z_1))(z_2 - m(z_2)) p_{t_1-t_2,0}(z_1, z_2) dz_1 dz_2$$

depends only on $t_1 - t_2$. Thus, $R(t_1, t_2) = R(t_1 - t_2)$.

Proof of properties of $R(t)$

Recall that $\mathrm{Var}(X) = \mathrm{Cov}(X, X)$ for any random variable X.

The case c), $|R(t)| \leq R(0)$ follows from

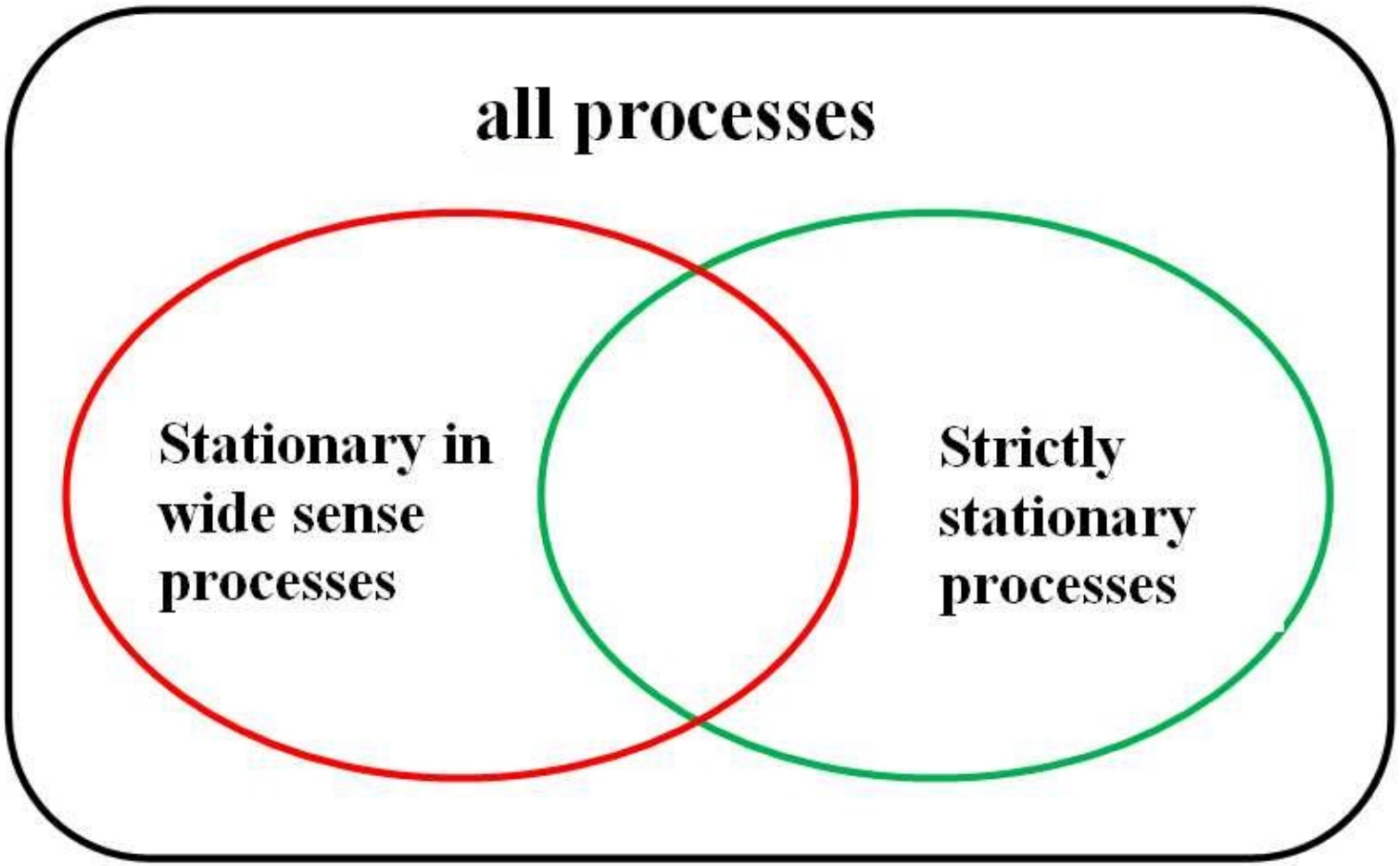
$$|\mathrm{Cov}(x_t, x_s)| \leq \sqrt{\mathrm{Var}(x_t)}\sqrt{\mathrm{Var}(x_s)},$$

which is proved in a probability theory course.

The case e) follows from

$$R(-t) = \mathrm{Cov}(x_{z-t}, x_z) = \mathrm{Cov}(x_z, x_{z-t}) = R(t).$$

Stationarity in strict and wide sense



Basic named processes

- Gaussian (normal) process
- White noise
- Random walk
- Brownian motion / Wiener process
- Brownian bridge
- Poisson process
- Subordinator
- Diffusion process
- Ornstein-Uhlenbeck (OU) process
- supOU process

and

- AR, MA, ARMA, ARIMA, SARIMA

Gaussian (normal) process

A stochastic process x_t, $t \in \mathbb{R}$, is called **Gaussian** (or normal) if for all $n, t_1, \ldots, t_n$ and all $c_1, \ldots, c_n$ the linear combination

$$\sum_{i=1}^{n} c_i x_{t_i}$$

has the Gaussian (normal) distribution.

Let $m(t) = \mathbf{E}x_t$ and $R(t,s) = \mathrm{Cov}(x_t, x_s)$.

Math task. Derive the 1- and 2-dimensional densities of the Gaussian process in terms of $m(t)$ and $R(t,s)$.

It follows from the definition that the vector $(x_{t_1}, \ldots, x_{t_n})$ has the multivariate normal distribution.

One-dimensional density of Gaussian process

For the Gaussian stochastic process x_t with continuous time, the 1-dimensional cumulative distribution function has the form

$$F_t(u) = P(x_t \leq u) = \int_{-\infty}^{u} p_t(z)dz,$$

where the 1-dimensional density is given by

$$p_t(z) = \frac{1}{\sqrt{2\pi R(t,t)}} e^{-\frac{(z-m(t))^2}{2R(t,t)}}.$$

Two-dimensional density of Gaussian process

For the Gaussian stochastic process x_t with continuous time, the 2-dimensional cumulative distribution function has the form

$$F_{t_1,t_2}(u_1,u_2)=P(x_{t_1}\le u_1, x_{t_2}\le u_2)=\int_{-\infty}^{u_1}\int_{-\infty}^{u_2} p_{t_1,t_2}(z_1,z_2)dz_2dz_1,$$

where the 2-dimensional density is defined by

$$p_{t_1,t_2}(z_1,z_2)=\frac{1}{\sqrt{(2\pi)^2\det\Sigma}}e^{-\frac{1}{2}v^T\Sigma^{-1}v},\ v=\begin{pmatrix} z_1-m(t_1)\\ z_2-m(t_2)\end{pmatrix},$$

$$\Sigma=\begin{pmatrix} R(t_1,t_1) & R(t_1,t_2)\\ R(t_2,t_1) & R(t_2,t_2)\end{pmatrix}.$$

Homework

Apply AI tools to deepen your understanding of the theoretical framework of stochastic processes.

- **Conceptual intuition:** Ask for a "plain English" comparison between *strict stationarity* and *wide sense stationarity*.
- **R simulations:** Use AI to generate code for simulating Gaussian processes and computing the densities.
- **Mathematical proofs:** Ask for step-by-step walkthroughs of properties like the symmetry of the autocovariance function $R(-t) = R(t)$.
- Solve exam-like exercises for L3, which are provided separately on Learning Central.